Topic 2A: Bonding	
1.	know that ionic bonding is the strong electrostatic attraction between oppositely charged ions
2.	understand the effects that ionic radius and ionic charge have on the strength of ionic bonding
3.	understand the formation of ions in terms of electron loss or gain
4.	be able to draw electronic configuration diagrams of cations and anions using dot-and-cross diagrams
5.	understand reasons for the trends in ionic radii down a group and for a set of isoelectronic ions, e.g. N^{3-} to Al^{3+}
6.	understand that the physical properties of ionic compounds and the migration of ions provide evidence for the existence of ions
7.	know that a covalent bond is the strong electrostatic attraction between two nuclei and the shared pair of electrons between them
8.	be able to draw dot-and-cross diagrams to show electrons in covalent substances, including: i molecules with single, double and triple bonds ii species exhibiting dative covalent (coordinate) bonding, including Al_2Cl_6 and ammonium ion
9.	understand the relationship between bond lengths and bond strengths for covalent bonds
10.	understand that the shape of a simple molecule or ion is determined by the repulsion between the electron pairs that surround a central atom
11.	understand reasons for the shapes of, and bond angles in, simple molecules and ions with up to six outer pairs of electrons (any combination of bonding pairs and lone pairs) *Examples should include $BeCl_2$, BCl_3, CH_4, NH_3, NH_4^+, H_2O, CO_2, $PCl_5(g)$ and $SF_6(g)$ and related molecules and ions; as well as simple organic molecules in this specification.*

Students should:
12. be able to predict the shapes of, and bond angles in, simple molecules and ions analogous to those specified above using electron-pair repulsion theory
13. know that electronegativity is the ability of an atom to attract the bonding electrons in a covalent bond
14. know that ionic and covalent bonding are the extremes of a continuum of bonding type and that electronegativity differences lead to bond polarity in bonds and molecules
15. understand that molecules with polar bonds may not be polar molecules and be able to predict whether or not a given molecule is likely to be polar
16. understand the nature of intermolecular forces resulting from the following interactions: i London forces (instantaneous dipole – induced dipole) ii permanent dipoles iii hydrogen bonds
17. understand the interactions in molecules, such as H_2O, liquid NH_3 and liquid HF, which give rise to hydrogen bonding
18. understand the following anomalous properties of water resulting from hydrogen bonding: i its relatively high melting temperature and boiling temperature ii the density of ice compared to that of water
19. be able to predict the presence of hydrogen bonding in molecules analogous to those mentioned above
20. understand, in terms of intermolecular forces, physical properties shown by materials, including: i the trends in boiling temperatures of alkanes with increasing chain length ii the effect of branching in the carbon chain on the boiling temperatures of alkanes iii the relatively low volatility (higher boiling temperatures) of alcohols compared to alkanes with a similar number of electrons iv the trends in boiling temperatures of the hydrogen halides, HF to HI
21. understand factors that influence the choice of solvents, including: i water, to dissolve some ionic compounds, in terms of the hydration of the ions ii water, to dissolve simple alcohols, in terms of hydrogen bonding iii water, as a poor solvent for compounds (to include polar molecules such as halogenoalkanes), in terms of inability to form hydrogen bonds iv non-aqueous solvents, for compounds that have similar intermolecular forces to those in the solvent
22. know that metallic bonding is the strong electrostatic attraction between metal ions and the delocalised electrons